

RETIE, Belgium

WMO: 064640

Lat: 51.2214N

Lon: 5.0272E

Elev: 21

StdP: 101.07

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-4.3	-9.5	1.7	-5.4	-6.5	2.2	-3.0	9.8	11.2	8.8	10.0	2.1	70	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	30.2	21.4	27.6	20.0	25.8	19.1	22.7	27.8	21.3	25.7	20.1	24.3	2.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	15.8	24.9	19.8	14.5	23.3	18.7	13.5	21.8	67.2	27.7	61.9	25.5	57.7	24.4	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.5	5.7	-9.9	34.7	4.4	2.1	-13.1	36.2	-15.6	37.5	-18.1	38.7	-21.3	40.2		
			-10.1	24.6	4.4	1.9	-13.2	26.0	-15.8	27.1	-18.2	28.1	-21.4	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	3.6	3.9	6.9	10.3	13.5	16.9	18.8	17.9	15.0	12.1	7.3	4.5	(6)
(7)		DBStd	6.44	4.04	4.22	3.63	3.74	3.80	3.08	3.28	2.72	2.77	3.54	3.40	3.93	(7)
(8)		HDD10.0	818	201	172	107	43	8	0	0	1	20	94	173		(8)
(9)		HDD18.3	2814	458	404	356	242	157	64	33	40	105	195	332	429	(9)
(10)		CDD10.0	1157	2	1	10	51	118	207	273	246	152	84	13	2	(10)
(11)		CDD18.3	112	0	0	0	1	8	21	48	27	6	1	0	0	(11)
(12)		CDH23.3	1140	0	0	0	18	79	218	483	270	63	8	0	0	(12)
(13)		CDH26.7	329	0	0	0	1	11	54	163	86	14	0	0	0	(13)
(14)	Wind	WSAvg	2.5	3.2	3.2	3.0	2.6	2.3	2.1	2.0	2.0	2.4	2.7	3.3	(14)	
(15)	Precipitation	PrecAvg	809	68	66	51	45	60	74	79	62	66	75	82	(15)	
(16)		PrecMax	1033	141	151	107	86	115	215	150	173	170	182	129	156	(16)
(17)		PrecMin	613	2	12	2	3	21	15	27	26	13	22	14	27	(17)
(18)		PrecStd	98	33	32	27	21	26	39	35	45	41	30	28	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	15.6	20.0	25.4	28.0	31.2	34.2	32.6	28.4	24.6	17.3	13.6	(19)
(20)		0.4%	MCWB	12.3	9.3	13.1	15.7	19.3	21.5	22.6	22.1	20.9	17.8	14.8	11.8	(20)
(21)		2%	DB	11.9	11.7	17.4	23.0	25.5	27.9	30.9	29.0	24.8	21.5	15.1	12.4	(21)
(22)		2%	MCWB	11.0	9.4	12.1	14.9	17.6	19.9	22.0	20.9	19.0	17.6	13.8	11.3	(22)
(23)		5%	DB	10.4	10.7	14.9	20.2	23.3	25.8	27.9	26.2	22.5	18.8	13.7	10.9	(23)
(24)		5%	MCWB	9.5	9.2	11.1	14.2	16.6	19.1	20.4	19.8	18.1	16.1	12.5	10.0	(24)
(25)		10%	DB	9.1	9.6	12.6	17.2	20.9	23.6	25.9	23.8	20.3	17.1	12.3	9.6	(25)
(26)		10%	MCWB	8.5	8.4	9.8	12.6	15.5	18.0	19.3	18.5	16.7	15.2	11.3	8.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	11.1	14.0	17.2	21.2	23.6	24.3	24.1	22.4	19.7	15.3	12.7	(27)
(28)		0.4%	MCDWB	13.1	12.4	18.1	22.8	26.0	29.0	30.4	29.0	26.5	23.1	16.4	13.1	(28)
(29)		2%	WB	11.1	10.1	12.9	15.8	18.9	21.5	22.9	22.0	20.0	17.7	14.0	11.4	(29)
(30)		2%	MCDWB	11.7	11.3	16.7	21.2	23.9	25.7	28.7	26.6	23.5	20.2	15.0	12.1	(30)
(31)		5%	WB	9.6	9.4	11.5	14.5	17.4	19.9	21.5	20.5	18.7	16.5	12.8	10.2	(31)
(32)		5%	MCDWB	10.2	10.6	14.3	19.0	21.6	24.1	26.4	24.5	21.4	18.2	13.5	10.9	(32)
(33)		10%	WB	8.5	8.4	10.3	13.2	16.0	18.6	20.2	19.2	17.4	15.6	11.4	9.0	(33)
(34)		10%	MCDWB	9.1	9.6	12.3	17.1	19.8	22.7	24.7	22.6	19.8	17.1	12.2	9.6	(34)
(35)	Mean Daily Temperature Range		MDBR	4.9	6.3	8.6	10.5	10.0	10.1	10.6	10.3	9.8	7.6	5.7	4.8	(35)
(36)		5% DB	MCDBR	5.4	7.7	13.0	14.9	13.7	14.1	15.1	14.7	13.1	10.0	6.4	5.3	(36)
(37)			MCWBR	4.8	5.7	8.3	7.9	7.0	6.9	6.8	7.3	7.7	6.4	5.3	4.7	(37)
(38)			5% WB	MCDBR	5.4	7.0	11.5	13.2	11.8	12.1	13.3	12.5	11.5	8.8	6.2	5.1
(39)		MCWBR		4.9	5.5	7.8	7.5	6.9	6.7	6.7	7.2	7.6	6.2	5.4	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.358	0.392	0.422	0.431	0.428	0.435	0.425	0.397	0.371	0.347	0.332	(40)	
(41)		taud	2.358	2.320	2.229	2.179	2.201	2.229	2.214	2.262	2.329	2.380	2.367	2.329	(41)	
(42)		Ebn at Noon	672	755	794	814	825	830	816	804	784	731	647	605	(42)	
(43)		Edh at Noon	65	86	113	133	136	134	133	121	102	81	64	57	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.76	1.44	2.60	4.16	4.98	5.39	5.08	4.33	3.22	1.89	0.89	0.58	(44)	
(45)		RadStd	0.10	0.28	0.37	0.46	0.44	0.45	0.61	0.41	0.31	0.15	0.09	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (85 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page